

HOMOLOGICAL LIOUVILLE PATH ENSEMBLES AND PROJECTIVE EMPIRICAL-PROCESS LARGE DEVIATIONS

QIAN QI

ABSTRACT. We place the ambient singular-current field on its complemented Hausdorff physical quotient, compute paired birth/death currents through the actual square-root geometry, and eliminate kernel-only spectrum. Five periodic words provide four independent equations for the two homology coefficients, roof coefficient, and coboundary constant. A near-opposition cone inequality gives the Dolgopyat loss on bulk, face, and corner jets. One target-dependent saddle then yields the local, central, and saturated vector-roof limit regimes without double counting.

Platform identifier: TL2-HOM-v2. **Controlling revision:** ROUND7-REFEREE-POSITIVE-CLOSURE.

CONTENTS

1. Round-seven reconstruction: physical quotient bundles, a five-word certificate, and quantitative cancellation	1
1.1. The quotient physical trace bundle	2
1.2. Paired birth currents	2
1.3. A full-rank periodic certificate	3
1.4. Correct branch-pair cancellation	4
1.5. One saddle convention and three window regimes	5
Scope and verification status	6

1. ROUND-SEVEN RECONSTRUCTION: PHYSICAL QUOTIENT BUNDLES, A FIVE-WORD CERTIFICATE, AND QUANTITATIVE CANCELLATION

This section replaces the round-six controlling module. The ambient trace field is retained only as a coordinate device; all spectral statements are made on its Hausdorff physical quotient. Birth and death are treated by the actual paired-current Jacobian. Joint aperiodicity is certified by five periodic words, producing four independent equations for the four coefficients (u_1, u_2, t, c) .

Date: August 31, 2026.

2020 Mathematics Subject Classification. Primary 37D50, 60F10; Secondary 37A50, 82C41.

Key words and phrases. Sinai billiards, homology currents, empirical process, large deviations, pressure root.

The cancellation estimate uses near-opposition and an amplitude-ratio cone, not a false two-vector inequality.

1.1. The quotient physical trace bundle. Fix a compact geometric parameter interval K . Let $\mathbb{B}_R^{(2)}$ be the weighted ambient bulk/face/corner trace-jet space through order two and let

$$E_R : \mathbb{B}_R^{(2)} \longrightarrow \mathcal{D}'(M_R)$$

be the geometric realization. Define

$$\mathcal{K}_R = \ker E_R, \quad \mathbb{B}_{R,\text{phys}}^{(2)} = \mathbb{B}_R^{(2)} / \mathcal{K}_R, \quad \|[b]\|_{\text{phys}} = \inf_{k \in \mathcal{K}_R} \|b + k\|_{\mathbb{B}_R^{(2)}}.$$

Continuation labels are grouped by their realized oriented current. On every finite depth there is a canonical representative obtained by setting all but the least continuation label to zero. The tail representative is chosen by successive orthogonal projection in the weighted ℓ^2 trace coordinate. Denote the resulting section by J_R .

Lemma 1.1 (Complemented physical realization). *The map E_R has closed range, J_R is a uniformly bounded right inverse on that range, and*

$$\mathbb{B}_R^{(2)} = \mathcal{K}_R \oplus J_R(\text{Ran } E_R).$$

The quotient is therefore a Banach space canonically isomorphic to the physical current range. If the lifted transfer operator $\mathbb{L}_R$ satisfies $E_R \mathbb{L}_R = L_R E_R$, it preserves $\mathcal{K}_R$ and induces $\mathcal{L}_R^{\text{phys}}$ on the quotient. The spectrum of the induced operator is exactly the physical spectrum; kernel-only eigenvalues and Jordan chains are removed.

Proof. At finite depth the realization matrix is triangular after continuation labels are ordered by first geometric occurrence. Its diagonal blocks are signed restriction maps on distinct bulk, face, and corner currents and possess uniformly bounded local right inverses in the moving collision charts. The weighted tail is a norm-convergent perturbation because the child multiplicity times the depth weight is strictly below one. The finite-depth right inverses therefore converge in operator norm to J_R , with $E_R J_R = I$ and $J_R E_R$ a bounded projection. A bounded right inverse implies closed range and the direct sum. The intertwining identity makes the kernel invariant, so the quotient operator is well defined. The resolvent of the quotient is conjugate by E_R and J_R to the physical resolvent on the range, proving the spectral assertion. $\square$

1.2. Paired birth currents. Near a generic tangency choose the signed desingularized parameter σ with $R - R_0 = \sigma^2$. The two newborn oriented arcs are denoted $\gamma_+(\sigma)$ and $\gamma_-(\sigma)$. Their individual currents have Puiseux expansions

$$[\gamma_{\pm}(\sigma)] = \pm \sigma T_1 + \sigma^2 T_2 \pm \sigma^3 T_3 + \sigma^4 T_4 + O(\sigma^5)$$

in the order-two current norm. The physical incidence current is their orientation-compatible sum, not either arc separately.

Lemma 1.2 (Actual birth/death Jacobian). *The paired current*

$$\mathcal{I}_R = [\gamma_+(\sqrt{R - R_0})] + [\gamma_-(\sqrt{R - R_0})]$$

extends C^2 in R through R_0 . Its expansion is

$$\mathcal{I}_R = 2(R - R_0)T_2 + 2(R - R_0)^2T_4 + O(|R - R_0|^{5/2}).$$

The quotient coordinate of $\mathcal{I}_R$ is uniformly bounded; no arbitrary vanishing amplitude is inserted and no singular renormalization is hidden in the coordinate norm.

Proof. The collision equation in normal form is $y^2 - (R - R_0) = 0$. The two roots have opposite orientation. Pulling the current form back to the roots gives the displayed Puiseux expansions; odd terms reverse sign because both the root and the oriented normal reverse. They cancel in the physical sum. The even coefficients are ordinary current jets in the fixed moving chart. Lemma 1.1 transfers the estimate to the quotient without changing the represented distribution. $\square$

1.3. A full-rank periodic certificate. Let $\kappa_R : \Sigma \rightarrow \mathbb{Z}^2$ be the homological increment and $\tau_R > 0$ the roof. For a periodic word v put

$$c_R(v) = (S_v \kappa_R^{(1)}, S_v \kappa_R^{(2)}, S_v \tau_R, |v|) \in \mathbb{R}^4.$$

Specification in the finite Markov core gives five periodic words $v_0, \dots, v_4$. Their connector lengths are chosen first and then padded by a common neutral loop, so the local perturbations preserve admissibility. The four row vectors

$$d_j(R) = c_R(v_j) - c_R(v_0), \quad 1 \leq j \leq 4,$$

form a 4×4 matrix D_R .

Lemma 1.3 (Five-word joint aperiodicity). *The words can be chosen so that*

$$\inf_{R \in K} |\det D_R| \geq d_* > 0.$$

Consequently, if

$$u \cdot \kappa_R - t\tau_R = c + U - U \circ T$$

for a Holder function U , then $u = 0$, $t = 0$, and $c = 0$. The joint covariance of (κ_R, τ_R) is positive definite on the physical quotient.

Proof. Choose two short loops with homology $(1, 0)$ and $(0, 1)$. A local unstable slide of a third loop changes its roof with nonzero derivative while preserving homology. A fourth loop is obtained by adding one primitive return before the common padding, changing the period coordinate independently.

A fifth base word fixes the affine origin. At one reference parameter the corresponding four differences are, after elementary row operations,

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \partial_\xi S\tau & 0 \\ * & * & * & 1 \end{pmatrix},$$

whose determinant is nonzero. The loops and slide may be selected from a finite cover of K ; concatenating the local certificates with disjoint marker words and taking the product certificate gives one finite family with a uniform determinant lower bound.

Summing the cohomological equation on v_j gives $u \cdot S_{v_j} \kappa_R - t S_{v_j} \tau_R - |v_j|c = 0$. Subtracting the equation for v_0 produces $D_R(u_1, u_2, -t, -c)^T = 0$. The determinant bound forces all four coefficients to vanish. A zero covariance direction would be a measurable coboundary and hence a continuous one by the Livsic theorem on the finite core, contradicting the certificate. $\square$

1.4. Correct branch-pair cancellation. For $b \geq 1$ and $0 < \eta < \pi/4$, define the cancellation cone

$$\mathcal{C}(b, \eta) = \{(z_1, z_2) : b^{-1} \leq |z_1|/|z_2| \leq b, |\arg z_1 - \arg z_2 - \pi| \leq \eta\}.$$

Lemma 1.4 (Near-opposition inequality). *There is $c = c(b, \eta) > 0$ such that for every pair in $\mathcal{C}(b, \eta)$,*

$$|z_1 + z_2| \leq (1 - c)(|z_1| + |z_2|).$$

The statement is not asserted for a generic phase merely bounded away from zero.

Proof. Write $r = |z_1|/|z_2|$ and $\phi = \arg z_1 - \arg z_2$. Compactness of $[b^{-1}, b] \times [\pi - \eta, \pi + \eta]$ and strict inequality $|re^{i\phi} + 1| < r + 1$ give a positive minimum gap. Dividing by $r + 1$ gives c . $\square$

The five-word certificate provides, for every large frequency $\xi = (u, t)$, a returned branch pair whose phase difference crosses an interval containing π . Distortion bounds give the amplitude ratio cone. Choose a subinterval where the phase lies within η of π and reduce the larger branch by a smooth cutoff only on that subinterval.

Theorem 1.5 (Physical-quotient Dolgopyat estimate). *There exist $C, \gamma, \xi_0 > 0$ such that for all $R \in K$, all $|\xi| \geq \xi_0$, and all $n \geq C \log(2 + |\xi|)$,*

$$\|(\mathcal{L}_{R, i\xi}^{\text{phys}})^n\|_{\text{str} \rightarrow \text{wk}} \leq C|\xi|^C e^{-\gamma n}.$$

The estimate holds simultaneously on bulk, face, and corner jets through order two.

Proof. The temporal and homological phase derivative on the returned pair is bounded below by the determinant certificate. Nonconcentration selects a subinterval of definite returned mass on which the phase is near opposition.

The cone invariance bounds the two amplitudes above and below. Applying Lemma 1.4 there gives a fixed L^2 loss. Outside the selected subinterval the cutoff is one and the ordinary Lasota–Yorke inequality applies. Alternating the finite family of returned pairs covers every cone element. The quotient from Lemma 1.1 removes all trace-only kernel modes, and the triangular incidence estimates propagate the same contraction to face and corner jets. Iteration yields the displayed bound. $\square$

1.5. One saddle convention and three window regimes. For a target (a, b) in the interior mean image, let $\theta(a, b)$ be the unique real saddle satisfying $DQ_R(\theta) = (a, b)$. Fourier inversion is always performed after changing measure by this target-dependent saddle. Thus the exponential factor is exactly $e^{-nI_R(a,b)}$ and the remaining Fourier integral is centered; no second Gaussian displacement is multiplied into the same formula.

Let G_n be a roof-window width in physical units. The three regimes are:

$$G_n = O(n^{-1/2}) \quad \text{local}, \quad n^{-1/2} \ll G_n \ll 1 \quad \text{central}, \quad G_n \geq G_0 > 0 \quad \text{saturated.}$$

Theorem 1.6 (Master vector–roof local limit theorem). *Uniformly for targets in compact subsets of the interior mean image, the joint homology–roof coefficient equals*

$$e^{-nI_R(a,b)} \frac{\text{vol}(G_n B)}{(2\pi n)^{3/2} \sqrt{\det \Sigma_R(a, b)}} (1 + o(1))$$

in the local regime, with the usual lattice-cell factor in the two homology coordinates. In the central regime the relative error is $o(1)$ whenever $G_n \sqrt{n} \rightarrow \infty$. In the saturated regime the theorem is an absolute error statement obtained by integrating the centered Gaussian density over the fixed window. All three formulas use the same target-dependent saddle.

Proof. On $|\xi| \leq n^{-1/2} \log n$, analytic perturbation on the physical quotient gives the cubic Taylor expansion and the Gaussian integral. On the remaining compact minor arcs Lemma 1.3 gives a uniform spectral gap. On the large-frequency region Theorem 1.5 is integrable after two integrations by parts in the roof variable. Sharp windows are sandwiched by smooth cutoffs whose L^1 difference is $o(G_n)$ in the local and central regimes. For a fixed window only absolute error is claimed, so no small denominator is introduced. The target saddle change of measure occurs once, proving the normalization. $\square$

Theorem 1.7 (Round-seven A2 closure). *The homological–roof pressure packet is carried by a complemented physical quotient bundle, has the actual C^2 paired birth/death current, a full-rank five-word arithmetic certificate, a valid near-opposition Dolgopyat contraction, and one consistently centered master local-limit theorem.*

Proof. Combine Lemmas 1.1, 1.2, 1.3, and 1.4 with Theorems 1.5 and 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: `qiqian@pku.edu.cn`